

For each problem, find all points of relative minima and maxima.

3) $y = -x^3 - 3x^2 - 1$

4) $y = x^4 - 2x^2 + 3$

5) $y = x^4 - x^2$

6) $y = -\frac{2}{x^2 - 4}$

7) $y = (2x - 8)^{\frac{2}{3}}$

8) $y = -\frac{1}{5}(x - 4)^{\frac{5}{3}} - 2(x - 4)^{\frac{2}{3}}$

Critical thinking questions:

9) Give an example function $f(x)$ where $f''(0) = 0$ and there is no relative minimum or maximum at $x = 0$.

10) Give an example function $f(x)$ where $f''(0) = 0$ and there is a relative maximum at $x = 0$.